1. understand that digestion breaks down natural polymers to smaller, soluble compounds that are absorbed and transported in the blood (illustrated by the breakdown of starch to glucose sugar and proteins to amino acids);
2. recall that cells grow by building up amino acids from the blood into new proteins;
3. recall that these parts of the body consist mainly of protein: muscle, tendons, skin, hair, haemoglobin in blood;
4. recall that excess amino acids are broken down in the liver to form urea, which is excreted by the kidneys in urine;
5. understand that high levels of sugar, common in some processed foods, are quickly absorbed into the blood stream, causing a rapid rise in the blood sugar level;
6. recall that there are two types of diabetes (type 1 and type 2), and that it is particularly late-onset diabetes (type 2) which is more likely to arise because of poor diet;
7. understand that obesity is one of the risk factors for type 2 diabetes;
8. understand that type 1 diabetes arises when the pancreas stops producing enough of the hormone, insulin: but that type 2 diabetes develops when the body no longer responds to its own insulin or does not make enough insulin;
9. recall that type 1 diabetes is controlled by insulin injections and that type 2 diabetes can be controlled by diet and exercise;
10. In the context of diet and health:
    - **can discuss a given risk, taking account of both the chance of it occurring and the consequences if it did;**
    - can suggest benefits of activities that have a known risk;
    - can offer reasons for people’s willingness (or reluctance) to accept the risk of a given activity;
    - can discuss personal and social choices in terms of a balance of risk and benefit.

This module features a central theme of modern chemistry. It shows how theories of atomic structure can be used to explain the properties of elements and their compounds. The module also includes examples to show how spectra and spectroscopy have contributed to the development of chemical knowledge and techniques. This module shows how atomic structure can be used to help explain the behaviour of elements.

The first topic looks at the Periodic Table and patterns that exist within it, focusing on Group 1 and Group 7. This topic also introduces the use of symbols and equations as a means of describing a chemical reaction. An explanation of the patterns is then developed in the next topic by linking atomic structure with chemical properties.

The third, and final, topic takes this further by introducing ions and showing how ionic theory can account for properties of compounds of Group 1 with Group 7 elements.

Topics
C4.1 What are the patterns in the properties of elements?
Classifying elements by their position in the Periodic Table; patterns in Group 1; patterns in Group 7; using symbols and equations to represent chemical reactions.
C4.2 How do chemists explain the patterns in the properties of elements?
Flame tests and spectra and their use for identifying elements and studying atomic structure. Classifying elements by their atomic structure; linking atomic structure to chemical properties.
C4.3 How do chemists explain the properties of compounds of Group 1 and Group 7 elements?
Ions; linking ion formation to atomic structure; Properties of ionic compounds of alkali metals and halogens.

ICT Opportunities

This module offers opportunities for illustrating the use of ICT in science, for example:

- storing large sets of data;
- selecting and presenting data in a variety of forms to explore patterns and trends.

Use of ICT in teaching and learning can include:

- an interactive periodic table to explore similarities and differences between elements;
- a spreadsheet to display patterns in chemical data;
- video clips to test predictions about the reactions of elements such as caesium and fluorine;
- the internet to research the uses of alkali metals or halogens and their compounds.